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Filtering sensitive topic speech within a conference audio stream

Abstract

A portion of an audio stream representing speech of a conference participant is filtered based on a determination that the portion of the audio stream corresponds to a predefined sensitive topic. An audio stream representing speech of a user of a participant device connected to a conference is obtained. First hash values are determined for portions of the speech. A determination is made, by comparing the first hash values against second hash values for records of a data store, that a portion of the speech corresponds to a predefined sensitive topic indicated within the records. A filter is applied against the portion of the speech to produce a modified audio stream within which the predefined sensitive topic is sanitized. An output, within the conference, of the modified audio stream is then caused in place of the audio stream.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8121845	12/2011	Kirby	N/A	N/A
8537978	12/2012	Jaiswal et al.	N/A	N/A
9413891	12/2015	Dwyer et al.	N/A	N/A
9443518	12/2015	Gauci	N/A	G06F 40/169
10432687	12/2018	Hanes et al.	N/A	N/A
10764534	12/2019	Shevchenko et al.	N/A	N/A
11450334	12/2021	Pichaimurthy et al.	N/A	N/A
11563855	12/2022	Spivak et al.	N/A	N/A
2004/0263636	12/2003	Cutler et al.	N/A	N/A
2007/0230372	12/2006	He	370/260	H04L 65/403
2013/0139259	12/2012	Tegreene	N/A	N/A
2013/0329866	12/2012	Mai et al.	N/A	N/A
2014/0028784	12/2013	Deyerle et al.	N/A	N/A
2015/0012270	12/2014	Reynolds	N/A	N/A
2015/0149173	12/2014	Korycki	N/A	N/A
2016/0063097	12/2015	Brown et al.	N/A	N/A
2022/0051652	12/2021	Winsvold et al.	N/A	N/A
2022/0199102	12/2021	Ostrand et al.	N/A	N/A
2023/0013497	12/2022	Aher et al.	N/A	N/A
2023/0117129	12/2022	Mouline et al.	N/A	N/A

OTHER PUBLICATIONS

MyFone, 6 Popular Real-Time Voice Changers for Zoom [2022 List], Karen William, Sep. 10, 2021 (Updated Jul. 5, 2022), 8 pages. cited by applicant

Voicemod, Voice Changer for Video Calls: ZOOM, Hangouts, Facetime, Sep. 2022, 2 pages. cited by applicant

Accent Conversion using Pre-trained Model and Synthesized Data from Voice Conversion, Tuan Nam Nguyen, Ngoc Quan Pham, Alexander Waibel, Karlsruhe Institute of Technology and Carnegie Mellon University, Sep. 2022, 5 pages. cited by applicant

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Background/Summary

FIELD

(1) This disclosure generally relates to audio stream modification, and, more specifically, to modifying an audio stream representing speech of a conference participant to filter a portion of the speech corresponding to a predefined sensitive topic.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) This disclosure is best understood from the following detailed description when read in conjunction with the accompanying drawings. It is emphasized that, according to common practice, the various features of the drawings are not to-scale. On the contrary, the dimensions of the various features are arbitrarily expanded or reduced for clarity.

(2) FIG. 1 is a block diagram of an example of an electronic computing and communications system.

(3) FIG. 2 is a block diagram of an example internal configuration of a computing device of an electronic computing and communications system.

(4) FIG. 3 is a block diagram of an example of a software platform implemented by an electronic computing and communications system.

(5) FIG. 4 is a block diagram of an example of a conferencing system for delivering conferencing software services in an electronic computing and communications system.

(6) FIG. 5 is a block diagram of an example of a conference participant audio stream modification system.

(7) FIG. 6 is a block diagram of an example of functionality of audio stream modification software.

(8) FIG. 7 is a block diagram of an example of a system for filtering sensitive topic speech within a conference audio stream.

(9) FIG. 8 is a block diagram of an example of a system for removing undesirable speech from within a conference audio stream.

(10) FIG. 9 is a flowchart of an example of a technique for filtering sensitive topic speech within a conference audio stream.

(11) FIG. 10 is a flowchart of an example of a technique for removing undesirable speech from within a conference audio stream.

DETAILED DESCRIPTION

(12) Conferencing software is frequently used across various industries to support video-enabled conferences between participants in multiple locations. In some cases, each of the conference participants separately connects to the conferencing software from their own remote locations. In other cases, one or more of the conference participants may be physically located in and connect to the conferencing software from a conference room or similar physical space (e.g., in an office setting) while other conference participants connect to the conferencing software from one or more remote locations. Conferencing software thus enables people to conduct video conferences without requiring them to be physically present with one another. Conferencing software may be available as a standalone software product or it may be integrated within a software platform, such as a unified communications as a service (UCaaS) platform.

(13) Participants to an audio or video conference may discuss a variety of topics during a conference. The topics may be pre-planned, such as by their inclusion within an agenda for the conference, or spontaneous, such as arising from the discussion of a different topic or upon request by a conference participant. In either case, a topic may from time to time be a sensitive topic, for example, information that is confidential or otherwise private. For example, where a conference includes participants associated with multiple, unrelated entities (e.g., two different companies as parties at some stage of a sales transaction), a participant from one of the entities may

unintentionally say something that is or relates to a topic which is sensitive to their entity and which should not be revealed to the participants of the other entity. This sensitive topic could, for example, be or otherwise relate to negotiation goals, project codenames, or financial details. However, conventional conferencing approaches do not have the capability to automatically detect references to such sensitive topics and modify audio from the subject participant to limit or prevent the exposure of those sensitive topic references to other participants within the conference.

(14) Participants to an audio or video conference may also use certain language or speech styles that differ from that of other participants, such as based on their personality, public speaking skills, or familiarity with a given topic. In some cases, speech from a conference participant may be considered to include undesirable language. For example, while some participants may be eloquent speakers, others may frequently and unintentionally use filler phrases such as “uh” or “um” that might have the effect of diluting an otherwise valuable thought to share or be seen as a disruption to an otherwise steady conversation. In another example, some participants may regularly use profanity in their personal lives and unintentionally use some in a professional conference setting, which may be considered unpleasant or offensive by others and thus may cause the speaker to lose standing within the conference. However, conventional conferencing approaches do not have the capability to automatically detect the use of such undesirable language and modify audio from the subject participant to limit or prevent the exposure of that language to other participants within the conference.

(15) Implementations of this disclosure address problems such as these by modifying audio streams representative of speech from a participant to an audio or video conference, in real-time during the conference and/or during a later playback of a recording of the conference. According to some implementations of this disclosure, a portion of an audio stream representing speech of a conference participant is filtered based on a determination that the portion of the audio stream corresponds to a predefined sensitive topic. An audio stream representing speech of a user of a participant device connected to a conference is obtained. Hash values are determined for portions of the speech. A determination is made, by comparing the hash values against hash values for records of a data store, that a portion of the speech corresponds to a predefined sensitive topic indicated within the records. A filter is applied against the portion of the speech to produce a modified audio stream within which the predefined sensitive topic is sanitized. An output, within the conference, of the modified audio stream is then caused in place of the audio stream. According to other implementations of this disclosure, a portion of speech represented in an audio stream of a conference participant is removed based on a determination that the portion of the speech corresponds to language identified as undesirable. An audio stream representing speech of a user of a participant device connected to a conference is obtained. The audio stream is processed to detect that a portion of the speech corresponds to language identified as undesirable within an audio profile. A modified audio stream is produced by removing the portion of the speech from the audio stream. An output, within the conference, of the modified audio stream is then played in place of the audio stream (e.g., by server-side or client-side software causing such output).

(16) To describe some implementations in greater detail, reference is first made to examples of hardware and software structures used to implement a conference participant audio stream modification system. FIG. 1 is a block diagram of an example of an electronic computing and communications system **100**, which can be or include a distributed computing system (e.g., a client-server computing system), a cloud computing system, a clustered computing system, or the like.

(17) The system **100** includes one or more customers, such as customers **102A** through **102B**, which may each be a public entity, private entity, or another corporate entity or individual that purchases or otherwise uses software services, such as of a UCaaS platform provider. Each customer can include one or more clients. For example, as shown and without limitation, the customer **102A** can include clients **104A** through **104B**, and the customer **102B** can include clients

104C through **104D**. A customer can include a customer network or domain. For example, and without limitation, the clients **104A** through **104B** can be associated or communicate with a customer network or domain for the customer **102A** and the clients **104C** through **104D** can be associated or communicate with a customer network or domain for the customer **102B**.

(18) A client, such as one of the clients **104A** through **104D**, may be or otherwise refer to one or both of a client device or a client application. Where a client is or refers to a client device, the client can comprise a computing system, which can include one or more computing devices, such as a mobile phone, a tablet computer, a laptop computer, a notebook computer, a desktop computer, or another suitable computing device or combination of computing devices. Where a client instead is or refers to a client application, the client can be an instance of software running on a customer device (e.g., a client device or another device). In some implementations, a client can be implemented as a single physical unit or as a combination of physical units. In some implementations, a single physical unit can include multiple clients.

(19) The system **100** can include a number of customers and/or clients or can have a configuration of customers or clients different from that generally illustrated in FIG. 1. For example, and without limitation, the system **100** can include hundreds or thousands of customers, and at least some of the customers can include or be associated with a number of clients.

(20) The system **100** includes a datacenter **106**, which may include one or more servers. The datacenter **106** can represent a geographic location, which can include a facility, where the one or more servers are located. The system **100** can include a number of datacenters and servers or can include a configuration of datacenters and servers different from that generally illustrated in FIG. 1. For example, and without limitation, the system **100** can include tens of datacenters, and at least some of the datacenters can include hundreds or another suitable number of servers. In some implementations, the datacenter **106** can be associated or communicate with one or more datacenter networks or domains, which can include domains other than the customer domains for the customers **102A** through **102B**.

(21) The datacenter **106** includes servers used for implementing software services of a UCaaS platform. The datacenter **106** as generally illustrated includes an application server **108**, a database server **110**, and a telephony server **112**. The servers **108** through **112** can each be a computing system, which can include one or more computing devices, such as a desktop computer, a server computer, or another computer capable of operating as a server, or a combination thereof. A suitable number of each of the servers **108** through **112** can be implemented at the datacenter **106**. The UCaaS platform uses a multi-tenant architecture in which installations or instantiations of the servers **108** through **112** is shared amongst the customers **102A** through **102B**.

(22) In some implementations, one or more of the servers **108** through **112** can be a non-hardware server implemented on a physical device, such as a hardware server. In some implementations, a combination of two or more of the application server **108**, the database server **110**, and the telephony server **112** can be implemented as a single hardware server or as a single non-hardware server implemented on a single hardware server. In some implementations, the datacenter **106** can include servers other than or in addition to the servers **108** through **112**, for example, a media server, a proxy server, or a web server.

(23) The application server **108** runs web-based software services deliverable to a client, such as one of the clients **104A** through **104D**. As described above, the software services may be of a UCaaS platform. For example, the application server **108** can implement all or a portion of a UCaaS platform, including conferencing software, messaging software, and/or other intra-party or inter-party communications software. The application server **108** may, for example, be or include a unitary Java Virtual Machine (JVM).

(24) In some implementations, the application server **108** can include an application node, which can be a process executed on the application server **108**. For example, and without limitation, the application node can be executed in order to deliver software services to a client, such as one of the

clients **104A** through **104D**, as part of a software application. The application node can be implemented using processing threads, virtual machine instantiations, or other computing features of the application server **108**. In some such implementations, the application server **108** can include a suitable number of application nodes, depending upon a system load or other characteristics associated with the application server **108**. For example, and without limitation, the application server **108** can include two or more nodes forming a node cluster. In some such implementations, the application nodes implemented on a single application server **108** can run on different hardware servers.

(25) The database server **110** stores, manages, or otherwise provides data for delivering software services of the application server **108** to a client, such as one of the clients **104A** through **104D**. In particular, the database server **110** may implement one or more databases, tables, or other information sources suitable for use with a software application implemented using the application server **108**. The database server **110** may include a data storage unit accessible by software executed on the application server **108**. A database implemented by the database server **110** may be a relational database management system (RDBMS), an object database, an XML database, a configuration management database (CMDB), a management information base (MIB), one or more flat files, other suitable non-transient storage mechanisms, or a combination thereof. The system **100** can include one or more database servers, in which each database server can include one, two, three, or another suitable number of databases configured as or comprising a suitable database type or combination thereof.

(26) In some implementations, one or more databases, tables, other suitable information sources, or portions or combinations thereof may be stored, managed, or otherwise provided by one or more of the elements of the system **100** other than the database server **110**, for example, the client **104** or the application server **108**.

(27) The telephony server **112** enables network-based telephony and web communications from and to clients of a customer, such as the clients **104A** through **104B** for the customer **102A** or the clients **104C** through **104D** for the customer **102B**. Some or all of the clients **104A** through **104D** may be voice over internet protocol (VOIP)-enabled devices configured to send and receive calls over a network **114**. In particular, the telephony server **112** includes a session initiation protocol (SIP) zone and a web zone. The SIP zone enables a client of a customer, such as the customer **102A** or **102B**, to send and receive calls over the network **114** using SIP requests and responses. The web zone integrates telephony data with the application server **108** to enable telephony-based traffic access to software services run by the application server **108**. Given the combined functionality of the SIP zone and the web zone, the telephony server **112** may be or include a cloud-based private branch exchange (PBX) system.

(28) The SIP zone receives telephony traffic from a client of a customer and directs same to a destination device. The SIP zone may include one or more call switches for routing the telephony traffic. For example, to route a VOIP call from a first VOIP-enabled client of a customer to a second VOIP-enabled client of the same customer, the telephony server **112** may initiate a SIP transaction between a first client and the second client using a PBX for the customer. However, in another example, to route a VOIP call from a VOIP-enabled client of a customer to a client or non-client device (e.g., a desktop phone which is not configured for VOIP communication) which is not VOIP-enabled, the telephony server **112** may initiate a SIP transaction via a VOIP gateway that transmits the SIP signal to a public switched telephone network (PSTN) system for outbound communication to the non-VOIP-enabled client or non-client phone. Hence, the telephony server **112** may include a PSTN system and may in some cases access an external PSTN system.

(29) The telephony server **112** includes one or more session border controllers (SBCs) for interfacing the SIP zone with one or more aspects external to the telephony server **112**. In particular, an SBC can act as an intermediary to transmit and receive SIP requests and responses between clients or non-client devices of a given customer with clients or non-client devices

external to that customer. When incoming telephony traffic for delivery to a client of a customer, such as one of the clients **104A** through **104D**, originating from outside the telephony server **112** is received, a SBC receives the traffic and forwards it to a call switch for routing to the client.

(30) In some implementations, the telephony server **112**, via the SIP zone, may enable one or more forms of peering to a carrier or customer premise. For example, Internet peering to a customer premise may be enabled to ease the migration of the customer from a legacy provider to a service provider operating the telephony server **112**. In another example, private peering to a customer premise may be enabled to leverage a private connection terminating at one end at the telephony server **112** and at the other end at a computing aspect of the customer environment. In yet another example, carrier peering may be enabled to leverage a connection of a peered carrier to the telephony server **112**.

(31) In some such implementations, a SBC or telephony gateway within the customer environment may operate as an intermediary between the SBC of the telephony server **112** and a PSTN for a peered carrier. When an external SBC is first registered with the telephony server **112**, a call from a client can be routed through the SBC to a load balancer of the SIP zone, which directs the traffic to a call switch of the telephony server **112**. Thereafter, the SBC may be configured to communicate directly with the call switch.

(32) The web zone receives telephony traffic from a client of a customer, via the SIP zone, and directs same to the application server **108** via one or more Domain Name System (DNS) resolutions. For example, a first DNS within the web zone may process a request received via the SIP zone and then deliver the processed request to a web service which connects to a second DNS at or otherwise associated with the application server **108**. Once the second DNS resolves the request, it is delivered to the destination service at the application server **108**. The web zone may also include a database for authenticating access to a software application for telephony traffic processed within the SIP zone, for example, a softphone.

(33) The clients **104A** through **104D** communicate with the servers **108** through **112** of the datacenter **106** via the network **114**. The network **114** can be or include, for example, the Internet, a local area network (LAN), a wide area network (WAN), a virtual private network (VPN), or another public or private means of electronic computer communication capable of transferring data between a client and one or more servers. In some implementations, a client can connect to the network **114** via a communal connection point, link, or path, or using a distinct connection point, link, or path. For example, a connection point, link, or path can be wired, wireless, use other communications technologies, or a combination thereof.

(34) The network **114**, the datacenter **106**, or another element, or combination of elements, of the system **100** can include network hardware such as routers, switches, other network devices, or combinations thereof. For example, the datacenter **106** can include a load balancer **116** for routing traffic from the network **114** to various servers associated with the datacenter **106**. The load balancer **116** can route, or direct, computing communications traffic, such as signals or messages, to respective elements of the datacenter **106**.

(35) For example, the load balancer **116** can operate as a proxy, or reverse proxy, for a service, such as a service provided to one or more remote clients, such as one or more of the clients **104A** through **104D**, by the application server **108**, the telephony server **112**, and/or another server. Routing functions of the load balancer **116** can be configured directly or via a DNS. The load balancer **116** can coordinate requests from remote clients and can simplify client access by masking the internal configuration of the datacenter **106** from the remote clients.

(36) In some implementations, the load balancer **116** can operate as a firewall, allowing or preventing communications based on configuration settings. Although the load balancer **116** is depicted in FIG. 1 as being within the datacenter **106**, in some implementations, the load balancer **116** can instead be located outside of the datacenter **106**, for example, when providing global routing for multiple datacenters. In some implementations, load balancers can be included both

within and outside of the datacenter **106**. In some implementations, the load balancer **116** can be omitted.

(37) FIG. 2 is a block diagram of an example internal configuration of a computing device **200** of an electronic computing and communications system. In one configuration, the computing device **200** may implement one or more of the client **104**, the application server **108**, the database server **110**, or the telephony server **112** of the system **100** shown in FIG. 1.

(38) The computing device **200** includes components or units, such as a processor **202**, a memory **204**, a bus **206**, a power source **208**, peripherals **210**, a user interface **212**, a network interface **214**, other suitable components, or a combination thereof. One or more of the memory **204**, the power source **208**, the peripherals **210**, the user interface **212**, or the network interface **214** can communicate with the processor **202** via the bus **206**.

(39) The processor **202** is a central processing unit, such as a microprocessor, and can include single or multiple processors having single or multiple processing cores. Alternatively, the processor **202** can include another type of device, or multiple devices, configured for manipulating or processing information. For example, the processor **202** can include multiple processors interconnected in one or more manners, including hardwired or networked. The operations of the processor **202** can be distributed across multiple devices or units that can be coupled directly or across a local area or other suitable type of network. The processor **202** can include a cache, or cache memory, for local storage of operating data or instructions.

(40) The memory **204** includes one or more memory components, which may each be volatile memory or non-volatile memory. For example, the volatile memory can be random access memory (RAM) (e.g., a DRAM module, such as DDR SDRAM). In another example, the non-volatile memory of the memory **204** can be a disk drive, a solid state drive, flash memory, or phase-change memory. In some implementations, the memory **204** can be distributed across multiple devices. For example, the memory **204** can include network-based memory or memory in multiple clients or servers performing the operations of those multiple devices.

(41) The memory **204** can include data for immediate access by the processor **202**. For example, the memory **204** can include executable instructions **216**, application data **218**, and an operating system **220**. The executable instructions **216** can include one or more application programs, which can be loaded or copied, in whole or in part, from non-volatile memory to volatile memory to be executed by the processor **202**. For example, the executable instructions **216** can include instructions for performing some or all of the techniques of this disclosure. The application data **218** can include user data, database data (e.g., database catalogs or dictionaries), or the like. In some implementations, the application data **218** can include functional programs, such as a web browser, a web server, a database server, another program, or a combination thereof. The operating system **220** can be, for example, Microsoft Windows®, Mac OS X®, or Linux®; an operating system for a mobile device, such as a smartphone or tablet device; or an operating system for a non-mobile device, such as a mainframe computer.

(42) The power source **208** provides power to the computing device **200**. For example, the power source **208** can be an interface to an external power distribution system. In another example, the power source **208** can be a battery, such as where the computing device **200** is a mobile device or is otherwise configured to operate independently of an external power distribution system. In some implementations, the computing device **200** may include or otherwise use multiple power sources. In some such implementations, the power source **208** can be a backup battery.

(43) The peripherals **210** includes one or more sensors, detectors, or other devices configured for monitoring the computing device **200** or the environment around the computing device **200**. For example, the peripherals **210** can include a geolocation component, such as a global positioning system location unit. In another example, the peripherals can include a temperature sensor for measuring temperatures of components of the computing device **200**, such as the processor **202**. In some implementations, the computing device **200** can omit the peripherals **210**.

- (44) The user interface **212** includes one or more input interfaces and/or output interfaces. An input interface may, for example, be a positional input device, such as a mouse, touchpad, touchscreen, or the like; a keyboard; or another suitable human or machine interface device. An output interface may, for example, be a display, such as a liquid crystal display, a cathode-ray tube, a light emitting diode display, or other suitable display.
- (45) The network interface **214** provides a connection or link to a network (e.g., the network **114** shown in FIG. **1**). The network interface **214** can be a wired network interface or a wireless network interface. The computing device **200** can communicate with other devices via the network interface **214** using one or more network protocols, such as using Ethernet, transmission control protocol (TCP), internet protocol (IP), power line communication, an IEEE 802.X protocol (e.g., Wi-Fi, Bluetooth, or ZigBee), infrared, visible light, general packet radio service (GPRS), global system for mobile communications (GSM), code-division multiple access (CDMA), Z-Wave, another protocol, or a combination thereof.
- (46) FIG. **3** is a block diagram of an example of a software platform **300** implemented by an electronic computing and communications system, for example, the system **100** shown in FIG. **1**. The software platform **300** is a UCaaS platform accessible by clients of a customer of a UCaaS platform provider, for example, the clients **104A** through **104B** of the customer **102A** or the clients **104C** through **104D** of the customer **102B** shown in FIG. **1**. The software platform **300** may be a multi-tenant platform instantiated using one or more servers at one or more datacenters including, for example, the application server **108**, the database server **110**, and the telephony server **112** of the datacenter **106** shown in FIG. **1**.
- (47) The software platform **300** includes software services accessible using one or more clients. For example, a customer **302** as shown includes four clients—a desk phone **304**, a computer **306**, a mobile device **308**, and a shared device **310**. The desk phone **304** is a desktop unit configured to at least send and receive calls and includes an input device for receiving a telephone number or extension to dial to and an output device for outputting audio and/or video for a call in progress. The computer **306** is a desktop, laptop, or tablet computer including an input device for receiving some form of user input and an output device for outputting information in an audio and/or visual format. The mobile device **308** is a smartphone, wearable device, or other mobile computing aspect including an input device for receiving some form of user input and an output device for outputting information in an audio and/or visual format. The desk phone **304**, the computer **306**, and the mobile device **308** may generally be considered personal devices configured for use by a single user. The shared device **310** is a desk phone, a computer, a mobile device, or a different device which may instead be configured for use by multiple specified or unspecified users.
- (48) Each of the clients **304** through **310** includes or runs on a computing device configured to access at least a portion of the software platform **300**. In some implementations, the customer **302** may include additional clients not shown. For example, the customer **302** may include multiple clients of one or more client types (e.g., multiple desk phones or multiple computers) and/or one or more clients of a client type not shown in FIG. **3** (e.g., wearable devices or televisions other than as shared devices). For example, the customer **302** may have tens or hundreds of desk phones, computers, mobile devices, and/or shared devices.
- (49) The software services of the software platform **300** generally relate to communications tools, but are in no way limited in scope. As shown, the software services of the software platform **300** include telephony software **312**, conferencing software **314**, messaging software **316**, and other software **318**. Some or all of the software **312** through **318** uses customer configurations **320** specific to the customer **302**. The customer configurations **320** may, for example, be data stored within a database or other data store at a database server, such as the database server **110** shown in FIG. **1**.
- (50) The telephony software **312** enables telephony traffic between ones of the clients **304** through **310** and other telephony-enabled devices, which may be other ones of the clients **304** through **310**,

other VOIP-enabled clients of the customer **302**, non-VOIP-enabled devices of the customer **302**, VOIP-enabled clients of another customer, non-VOIP-enabled devices of another customer, or other VOIP-enabled clients or non-VOIP-enabled devices. Calls sent or received using the telephony software **312** may, for example, be sent or received using the desk phone **304**, a softphone running on the computer **306**, a mobile application running on the mobile device **308**, or using the shared device **310** that includes telephony features.

(51) The telephony software **312** further enables phones that do not include a client application to connect to other software services of the software platform **300**. For example, the telephony software **312** may receive and process calls from phones not associated with the customer **302** to route that telephony traffic to one or more of the conferencing software **314**, the messaging software **316**, or the other software **318**.

(52) The conferencing software **314** enables audio, video, and/or other forms of conferences between multiple participants, such as to facilitate a conference between those participants. In some cases, the participants may all be physically present within a single location, for example, a conference room, in which the conferencing software **314** may facilitate a conference between only those participants and using one or more clients within the conference room. In some cases, one or more participants may be physically present within a single location and one or more other participants may be remote, in which the conferencing software **314** may facilitate a conference between all of those participants using one or more clients within the conference room and one or more remote clients. In some cases, the participants may all be remote, in which the conferencing software **314** may facilitate a conference between the participants using different clients for the participants. The conferencing software **314** can include functionality for hosting, presenting scheduling, joining, or otherwise participating in a conference. The conferencing software **314** may further include functionality for recording some or all of a conference and/or documenting a transcript for the conference.

(53) The messaging software **316** enables instant messaging, unified messaging, and other types of messaging communications between multiple devices, such as to facilitate a chat or other virtual conversation between users of those devices. The unified messaging functionality of the messaging software **316** may, for example, refer to email messaging which includes a voicemail transcription service delivered in email format.

(54) The other software **318** enables other functionality of the software platform **300**. Examples of the other software **318** include, but are not limited to, device management software, resource provisioning and deployment software, administrative software, third party integration software, and the like. In one particular example, the other software **318** can include software for modifying audio streams representing speech of participants to a conference within the conference. In some such cases, the other software **318** may be or be included in the conferencing software **314**.

(55) The software **312** through **318** may be implemented using one or more servers, for example, of a datacenter such as the datacenter **106** shown in FIG. 1. For example, one or more of the software **312** through **318** may be implemented using an application server, a database server, and/or a telephony server, such as the servers **108** through **112** shown in FIG. 1. In another example, one or more of the software **312** through **318** may be implemented using servers not shown in FIG. 1, for example, a meeting server, a web server, or another server. In yet another example, one or more of the software **312** through **318** may be implemented using one or more of the servers **108** through **112** and one or more other servers. The software **312** through **318** may be implemented by different servers or by the same server.

(56) Features of the software services of the software platform **300** may be integrated with one another to provide a unified experience for users. For example, the messaging software **316** may include a user interface element configured to initiate a call with another user of the customer **302**. In another example, the telephony software **312** may include functionality for elevating a telephone call to a conference. In yet another example, the conferencing software **314** may include

functionality for sending and receiving instant messages between participants and/or other users of the customer **302**. In yet another example, the conferencing software **314** may include functionality for file sharing between participants and/or other users of the customer **302**. In some implementations, some or all of the software **312** through **318** may be combined into a single software application run on clients of the customer, such as one or more of the clients **304** through **310**.

(57) FIG. **4** is a block diagram of an example of a conferencing system **400** for delivering conferencing software services in an electronic computing and communications system, for example, the system **100** shown in FIG. **1**. The conferencing system **400** includes a thread encoding tool **402**, a switching/routing tool **404**, and conferencing software **406**. The conferencing software **406**, which may, for example, be the conferencing software **314** shown in FIG. **3**, is software for implementing conferences (e.g., video conferences) between users of clients and/or phones, such as clients **408** and **410** and phone **412**. For example, the clients **408** or **410** may each be one of the clients **304** through **310** shown in FIG. **3** that runs a client application associated with the conferencing software **406**, and the phone **412** may be a telephone which does not run a client application associated with the conferencing software **406** or otherwise access a web application associated with the conferencing software **406**. The conferencing system **400** may in at least some cases be implemented using one or more servers of the system **100**, for example, the application server **108** shown in FIG. **1**. Although two clients and a phone are shown in FIG. **4**, other numbers of clients and/or other numbers of phones can connect to the conferencing system **400**.

(58) Implementing a conference includes transmitting and receiving video, audio, and/or other data between clients and/or phones, as applicable, of the conference participants. Each of the client **408**, the client **410**, and the phone **412** may connect through the conferencing system **400** using separate input streams to enable users thereof to participate in a conference together using the conferencing software **406**. The various channels used for establishing connections between the clients **408** and **410** and the phone **412** may, for example, be based on the individual device capabilities of the clients **408** and **410** and the phone **412**.

(59) The thread encoding tool **402** receives video streams separately from the clients **408** and **410** and encodes those video streams using one or more transcoding tools, such as to produce variant streams at different resolutions. For example, a given video stream received from a client may be processed using multi-stream capabilities of the conferencing system **400** to result in multiple resolution versions of that video stream, including versions at 90p, 180p, 360p, 720p, and/or 1080p, amongst others. The video streams may be received from the clients over a network, for example, the network **114** shown in FIG. **1**, or by a direct wired connection, such as using a universal serial bus (USB) connection or like coupling aspect. After the video streams are encoded, the switching/routing tool **404** directs the encoded streams through applicable network infrastructure and/or other hardware to deliver the encoded streams to the conferencing software **406**. The conferencing software **406** transmits the encoded video streams to each connected client, such as the clients **408** and **410**, which receive and decode the encoded video streams to output the video content thereof for display by video output components of the clients.

(60) A user of the phone **412** participates in a conference using an audio-only connection and may be referred to as an audio-only caller. To participate in the conference from the phone **412**, an audio stream from the phone **412** is received and processed at a VOIP gateway **414** to prepare a digital telephony signal for processing at the conferencing system **400**. The VOIP gateway **414** may be part of the system **100**, for example, implemented at or in connection with a server of the datacenter **106**, such as the telephony server **112** shown in FIG. **1**. Alternatively, the VOIP gateway **414** may be located on the user-side, such as in a same location as the phone **412**. The digital telephony signal is a packet switched signal transmitted to the switching/routing tool **404** for delivery to the conferencing software **406**. The conferencing software **406** outputs an audio stream representing a combined audio capture for each participant of the conference for output by an audio

output component of the phone **412**. In some implementations, the VOIP gateway **414** may be omitted, for example, where the phone **412** is a VOIP-enabled phone.

(61) A conference implemented using the conferencing software **406** may be referred to as a video conference in which video streaming is enabled for the conference participants thereof. The enabling of video streaming for a conference participant of a video conference does not require that the conference participant activate or otherwise use video functionality for participating in the video conference. For example, a conference may still be a video conference where none of the participants joining using clients turn on their video stream for any portion of the conference. In some cases, however, the conference may have video disabled, such as where each participant connects to the conference using a phone rather than a client, or where a host of the conference selectively configures the conference to exclude video functionality.

(62) In some implementations, other software services may be accessible in connection with a conference implemented using the conferencing system **400**. For example, a conference may include or otherwise integrate functionality for instant messaging, unified messaging, and other types of messaging communications between participants of the conference, such as to facilitate a chat or like virtual conversation between participants. Those other software services may be implemented at the conferencing system **400** and/or a different aspect of the system **100**.

(63) FIG. 5 is a block diagram of an example of a conference participant audio stream modification system **500**. The system **500** includes a server device **502**, a participant device **504**, and other participant devices **506**. Each of the participant device **504** and the other participant devices **506** may, for example, be one of the clients **408** or **410**. In some cases, one or more of the other participant devices **506** may be a phone, for example, the phone **412**. The participant device **504** and the other participant devices **506** are used to connect to and participate in an audio or video conference implemented by conferencing software **508** at the server device **502**. The conferencing software **508** may, for example, be the conferencing software **406**.

(64) The participant device **504** connects to the conferencing software **508** using a client application **510**, which is a client-side software application running at the participant device **504**. The client application **510** may, for example, be a desktop software application, mobile application, or web application associated with one or more services of a software platform, for example the software platform **300** shown in FIG. 3. The client application **510** may be software that allows a user of the participant device **504** to access or otherwise use one or more of the software **312** through **318** shown in FIG. 3. While not shown, at least some of the other participant devices **506** may also connect to the conference using a client application running at those ones of the other participant devices **506**.

(65) The client application **510** includes or otherwise uses audio stream modification software **512** for modifying an audio stream representing speech of the user of the participant device **504**. In particular, the audio stream modification software **512** is configured to automatically detect and remediate the use of certain contents within speech of a participant to a conference (e.g., the user of the participant device **504**). In one example, the audio stream modification software **512** automatically detects references to a sensitive topic within speech represented by an audio stream from a participant device (e.g., the participant device **504**) and modifies the audio stream to limit or prevent the exposure of that sensitive topic to other participants within the conference (e.g., users of the other participant devices **506**). In another example, the audio stream modification software **512** automatically detects the use of undesirable language within speech represented by an audio stream from a participant device (e.g., the participant device **504**) and modifies the audio stream to limit or prevent the exposure of that undesirable language to other participants within the conference (e.g., users of the other participant devices **506**). In either case, the automatic detection is in response to or otherwise based on the subject audio stream being obtained by the audio stream modification software **512**.

(66) References to an audio stream being modified and/or to a modified audio stream being output

by the audio stream modification software **512** “within a conference” are non-limiting as to a time at which such modification and/or output occurs. In particular, the audio stream modification software **512** may modify an audio stream and output the modified audio stream during a conference (i.e., while the conference remains on-going) or during playback of a recording of the conference (i.e., after the conference has ended).

(67) While the audio stream modification software **512** is shown and described as being included within the client application **510**, in some implementations, the audio stream modification software **512** or a portion thereof may be external to the client application **510**. For example, the audio stream modification software **512** may be implemented at the conferencing software **508**. In another example, the audio stream modification software **512** may in some cases represent functionality of a software platform which includes the conferencing software **508** (e.g., the software platform **300**) external to but otherwise for use with the conferencing software **508**.

(68) FIG. **6** is a block diagram of an example of functionality of the audio stream modification software **512**. The audio stream modification software **512** includes tools, such as programs, subprograms, functions, routines, subroutines, operations, and/or the like, for modifying audio streams representing speech of participants to a conference within the conference. As shown, the audio stream modification software **512** includes a sensitive topic detection tool **600**, an undesirable language detection tool **602**, a modified audio stream production tool **604**, and a modified audio stream replacement tool **606**.

(69) The sensitive topic detection tool **600** determines that a portion of speech represented within an audio stream from a participant device (e.g., the participant device on which the audio stream modification software **512** is running or a participant device connected to a server device on which the audio stream modification software **512** is running) corresponds to a predefined sensitive topic. The predefined sensitive topic is one of multiple predefined sensitive topics to which speech might correspond. The predefined sensitive topics are topics which have been identified as sensitive in nature before the conference in which the audio stream modification software **512** is used to modify a given audio stream, or at least before the subject audio stream modification occurs during the conference. Sensitive topics include confidential information, private information, secret information, or other information for which public disclosure is prohibited, restricted, or unwanted. Examples of sensitive topics include, without limitation, secretive business information (e.g., projects under internal development or sales figures), financial information (e.g., account balances or transaction references), and personal identity information (e.g., names or addresses). Topics may be identified as sensitive by or from one or more people or sources, including, without limitation, a user of a participant device, an administrator associated with an account or entity with which one or more participant devices are registered, or a person having other authority within that entity (e.g., a manager or supervisor).

(70) Records indicative of the predefined sensitive topics are maintained in a predefined sensitive topics data store accessible to the audio stream modification software **512**. The predefined sensitive topics data store may be under the control of an administrator associated with an account or entity with which one or more participant devices connected to a conference are associated. The administrator may, for example, add records to, modify records of, and/or delete records from the predefined sensitive topics data store according to changes in what topics are considered sensitive. For example, upon a new codename being created for a project under development by an entity, a record associated with that codename may be added to the predefined sensitive topics data store. In another example, upon a project associated with the codename being revealed to the public (e.g., during a product launch), a record associated with that codename may be removed from the predefined sensitive topics data store. In that sensitive topics are generally sensitive to multiple users (e.g., to an entire entity), the predefined sensitive topics data store may not be configured for control by individual users of participant devices.

(71) A record within the predefined sensitive topics data store may include one or more hash values

representative of the topic with which the record is associated. In particular, each hash value may correspond to a different semantic expression of that topic. For example, a record within the predefined sensitive topics data store may correspond to a company's contractual agreement with a business partner, Acme Holdings. A first hash value for the record may be created based on the expression "the Acme Holdings agreement," a second hash value may be created based on the expression "the agreement with Acme Holdings," a third hash value may be created based on the expression "the Acme contract," and so on. A hash value may be created by hashing audio input or text input representative of the subject expression. The hashing is performed according to a hash function such as Secure Hash Algorithm (SHA)-1, SHA-2, or SHA-3. In some cases, different expressions for a given sensitive topic may be manually provided for hashing, such as by an administrator. In some cases, different expressions for a given sensitive topic may be determined using a semantic neighborhood process, for example, using a K-nearest neighbor search process modeled based on semantic meanings of various words and/or using a machine learning model trained for semantic neighbor evaluation.

(72) The sensitive topic detection tool **600** determines that a portion of speech represented within an audio stream from a participant device corresponds to a predefined sensitive topic by comparing hash values determined for portions of the speech represented within the audio stream against hash values for records of the predefined sensitive topics data store. In particular, upon obtaining an audio stream representing speech of a user of a participant device connected to a conference, the sensitive topic detection tool **600** determines first hash values for portions of the speech. A portion of the speech for which a first hash value is determined may be or include a single word, a combination of words, a partial phrase, a complete phrase, a partial sentence, or a complete sentence. The speech may, for example, be hashed in portions identified based on temporal chunks (e.g., 1 or 5 second intervals).

(73) The first hash values are compared against second hash values for various records of the predefined sensitive topics data store, in which the terms "first" and "second" are used merely to explain that the hash values are of separate sets rather than implying an ordering or other relationship. The comparison is to determine whether any second hash value for a record of the predefined sensitive topics data store matches any first hash value. In some cases, all hash values for all records of the predefined sensitive topics data store may be searched in some search order until the earlier to occur of a hash value matching one of the first hash values is identified or a determination is made that no hash value within the predefined sensitive topics data store matches any of the first hash values. Thus, where a match is found between one of the first hash values and one of the second hash values, the sensitive topic detection tool **600** outputs an indication of that match. In such a case, a sensitive topic is detected in the speech represented within the audio stream. However, where a match is not found between any of the first hash values and any of the second hash values, the sensitive topic detection tool **600** outputs an indication that no match was found. In such a case, a sensitive topic is not detected in the speech represented within the audio stream.

(74) The sensitive topic detection tool **600** is generally configured to determine that a portion of the speech represented within an audio stream corresponds to a predefined sensitive topic regardless of who the participants to the subject conference are. However, in some implementations, the sensitive topic detection tool **600** may be configured to determine that a portion of the speech represented within an audio stream corresponds to a predefined sensitive topic based on there being at least two different domains (e.g., domain names for registered email addresses) for user accounts associated with users of participant devices connected to a subject conference. For example, where all user accounts associated with the users of the participant devices are common to a same domain (e.g., "examplecompany.com"), the sensitive topic detection tool **600** may be disabled or otherwise unused based on the premise that there is no risk of exposure of sensitive topic information to others outside of that domain within the conference. In some implementations, the sensitive topic

detection tool **600** may be configured to determine that a portion of the speech represented within an audio stream corresponds to a predefined sensitive topic based on a social graph or organization chart for an entity with which the user of the participant device from which the audio stream is obtained is associated. For example, the sensitive topic detection tool **600** may be configured to determine that a portion of the speech represented within an audio stream corresponds to a predefined sensitive topic where an organization chart indicates that participants to the subject conference are from separate teams even within a single entity (i.e., such that the participants likely share a common domain) which operate under a strict policy against sharing certain work information.

(75) In some implementations, the sensitive topic detection tool **600** determines that a portion of speech represented within an audio stream from a participant device corresponds to a predefined sensitive topic other than based on a hash value comparison. For example, the sensitive topic detection tool **600** can perform text conversion to convert a portion of speech represented within an audio stream from a participant device into text. The sensitive topic detection tool **600** can then compare the text against text of records of the predefined sensitive topic data store, which each include text representative of various predefined sensitive topics (e.g., including various semantic meaning versions thereof). A determination that the portion of the speech corresponds to a predefined topic may thus be made based on a match between the text produced by the text conversion and text of a record of the predefined sensitive topic data store.

(76) The undesirable language detection tool **602** determines that a portion of speech represented within an audio stream from a participant device (e.g., the participant device on which the audio stream modification software **512** is running or a participant device connected to a server device on which the audio stream modification software **512** is running) corresponds to language identified as undesirable. In particular, the undesirable language detection tool **602** determines that the portion of the speech corresponds to language identified as undesirable within an audio profile. The audio profile is or includes records of language spoken by one or more participant device users in one or more past conferences. For example, the audio profile may correspond to the user of the participant device from which the audio stream is obtained, in which case the audio profile includes records identifying, as undesirable, language spoken by the user of the participant device in one or more past conferences. In another example, the audio profile may correspond to an entity with which the user of the participant device from which the audio stream is obtained is associated, in which case the audio profile includes records identifying, as undesirable, language spoken by one or more users of participant devices associated with the entity in one or more past conferences. Language may be identified or marked for identification as being undesirable by or from one or more people or sources, including, without limitation, a user of a participant device, an administrator associated with an account or entity with which one or more participant devices are registered, or a person having other authority within that entity (e.g., a manager or supervisor).

(77) Generally, undesirable language is language which is undesirable to speak in a professional conference setting. Examples of undesirable language may include, without limitation, profanity, filler words, or phrases which would be considered offensive to one or more groups of people. The quality of language being undesirable may be with reference to the speaker of the language or a recipient of the language, and thus language may be considered undesirable by a speaker and/or a recipient. Because the types of words or phrases which may qualify as undesirable language may differ between countries or regions or the world, the detection of undesirable language by the undesirable language detection tool **602** is based on vernacular rather than an assumed global meaning for certain words or phrases. The audio profile which is used to determine that portions of speech correspond to language identified as undesirable thus may be specific to a location in which the user or users of the participant device is or are located.

(78) In particular, the audio profile may identify language frequently used by the user or users in one or more past conferences along with an indication of whether the language is undesirable. In

some cases, the audio profile may identify language not used in a past conference but which is still considered undesirable, such as profanity, disparaging words or phrases, or words or phrases commonly recognized as relating to unwanted advances or innuendos. In some cases, the audio profile may only include language identified as undesirable. Language may be identified as undesirable within the audio profile based on manual user input indicating that the language is undesirable language. For example, a user of a participant device (i.e., the participant device from which the audio stream processed by the undesirable language detection tool **602** is processed or another participant device) may receive a prompt from the audio stream modification software **512** asking the user to verify whether a given word or phrase qualifies as undesirable language (e.g., based on the language being unpleasant, offensive, disruptive, or an annoyance). In another example, a machine learning model may be trained to recognize words or phrases which may qualify as undesirable language based on patterns extrapolated from a labeled training data set indicating words or phrases that do or do not qualify as undesirable language.

(79) Similar to the processing described above by the sensitive topic detection tool **600**, each element of language within the audio profile may include a hash value created by hashing audio input or text input representative of the subject language. The hashing is performed according to a hash function such as SHA-1, SHA-2, or SHA-3. The undesirable language detection tool **602** determines that a portion of speech represented within an audio stream from a participant device corresponds to language identified as undesirable within an audio profile by comparing hash values determined for portions of the speech represented within the audio stream against hash values for elements of language (i.e., different words or phrases) within the audio profile. In particular, upon obtaining an audio stream representing speech of a user of a participant device connected to a conference, the undesirable language detection tool **602** determines first hash values for portions of the speech. A portion of the speech for which a first hash value is determined may be or include a single word, a combination of words, a partial phrase, a complete phrase, a partial sentence, or a complete sentence. The speech may, for example, be hashed in portions identified based on temporal chunks (e.g., 1 or 5 second intervals).

(80) The first hash values are compared against second hash values for elements of language within the audio profile, in which, as with the sensitive topics described above, the terms “first” and “second” are used merely to explain that the hash values are of separate sets rather than implying an ordering or other relationship. The comparison is to determine whether any second hash value for language identified as undesirable within the audio profile matches any first hash value. In some cases, all hash values for all elements of language within the audio profile may be searched in some search order until the earlier to occur of a hash value matching one of the first hash values is identified or a determination is made that no hash value within the audio profile matches any of the first hash values. Thus, where a match is found between one of the first hash values and one of the second hash values, the undesirable language detection tool **602** outputs an indication of that match. In such a case, undesirable language is detected in the speech represented within the audio stream. However, where a match is not found between any of the first hash values and any of the second hash values, the undesirable language detection tool **602** outputs an indication that no match was found. In such a case, undesirable language is not detected in the speech represented within the audio stream.

(81) The undesirable language detection tool **602** is generally configured to determine that a portion of the speech represented within an audio stream corresponds to language identified as undesirable regardless of who the participants to the subject conference are. However, in some implementations, the undesirable language detection tool **602** may be configured to determine that a portion of the speech represented within an audio stream corresponds to language identified as undesirable based on there being at least two different domains (e.g., domain names for registered email addresses) for user accounts associated with users of participant devices connected to a subject conference. For example, where all user accounts associated with the users of the

participant devices are common to a same domain (e.g., “examplecompany.com”), the undesirable language detection tool **602** may be disabled or otherwise unused based on the premise that there is no risk of exposure of undesirable language to others outside of that domain within the conference. (82) In some implementations, the undesirable language detection tool **602** determines that a portion of speech represented within an audio stream from a participant device corresponds to language identified as undesirable other than based on a hash value comparison. For example, the undesirable language detection tool **602** can perform text conversion to convert a portion of speech represented within an audio stream from a participant device into text. The undesirable language detection tool **602** can then compare the text against text of an audio profile, in which the text of the audio profile represents various undesirable language (e.g., including various semantic meaning versions thereof). A determination that the portion of the speech corresponds to language identified as undesirable may thus be made based on a match between the text produced by the text conversion and text of the audio profile.

(83) In some implementations, the audio profile may omit language associated with one or more conference participant names that otherwise phonetically similar to undesirable language. For example, a conference participant may have a name which sounds like language otherwise identified as undesirable within the audio profile. In such a case, the participant device user associated with the audio profile may be prompted to verify whether the name, given its use in a past conference, should be identified as undesirable language within the audio profile. In another example, language added to the audio profile based on such a participant name may be automatically removed from the audio profile based on a determination that the name is phonetically similar to language otherwise recognized as undesirable.

(84) The modified audio stream production tool **604** produces a modified audio stream according to the output from the sensitive topic detection tool **600**, the undesirable language detection tool **602**, or both. In particular, the modified audio stream production tool **604** causes a filtering (e.g., a removal or obfuscation) of the portion of the speech detected, by the sensitive topic detection tool **600** or the undesirable language detection tool **602**, respectively, as corresponding to a predefined sensitive topic or to language identified as undesirable. For example, the modified audio stream production tool **604** can apply a filter against the portion of the speech to produce a modified audio stream within which the predefined sensitive topic or the undesirable language is filtered. In another example, the modified audio stream production tool **604** can produce a modified audio stream by removing the portion of the speech from the audio stream. In one non-limiting example, filtering the predefined sensitive topic or the undesirable language within or removing the predefined sensitive topic or the undesirable language from the audio stream can include obfuscating the portion of the speech. In such a case, the filter may be or include some configuration for distorting targeted portions of the speech.

(85) The modified audio stream replacement tool **606** causes an output, within the conference, of the modified audio stream produced by the modified audio stream production tool **606** in place of the audio stream originally obtained from the subject participant device. Generally, the modified audio stream is output at each participant device connected to the conference. However, in some implementations, the modified audio stream may be output at some, but not all, of the participant devices connected to the conference. For example, the modified audio stream may be output only at participant devices at which user accounts of the users have a domain different from a domain of the user account associated with the participant device from which the audio stream is obtained. In such a case, the modified audio stream may be output at a first group of participant devices while the original audio stream (i.e., without modification) may be output at a second group of participant devices.

(86) The operation of the audio stream modification software **512** is generally with respect to an ongoing conference. Thus, in the examples described above, the modification of an audio stream using the audio stream modification software **512** and thus the production of the modified audio

stream is described as occurring during the conference and therefore while the conference remains ongoing. However, in some implementations, the audio stream modification software **512** may operate against a recording of the playback and thus the modified audio stream may be produced during playback of a recording of the conference. In such a case, the audio stream modification is performed at a device at which the recording of the conference is being played back. In some such cases, the production of the modified audio stream and/or the processing by the sensitive topic detection tool **600** and/or the undesirable language detection tool **602** may be based on the device at which the recording of the conference is being played corresponding to a domain different from that of a participant device from which a given audio stream output during the recording playback was obtained within the conference.

(87) Although the tools **600** through **606** are shown as separate tools, in some implementations, two or more of the tools **600** through **606** may be combined into a single tool. Although the tools **600** through **606** are shown as functionality of the audio stream modification software **512** as a single piece of software, in some implementations, some or all of the tools **600** through **606** may exist outside of the audio stream modification software **512**. Similarly, in some implementations, a software service using the audio stream modification software **512** (e.g., the conferencing software **508**) may exclude the audio stream modification software **512** while still including the some or all of tools **600** through **606** in some form elsewhere or otherwise make use of the tools **600** through **606** while some or all of the tools **600** through **606** are included in some form elsewhere.

(88) While the audio stream modification software **512** is shown and described herein as being run at a participant device (e.g., as part of the client application **510**), in some implementations, the audio stream modification software **512** or a portion thereof may be run other than at a participant device. For example, the audio stream modification software **512** may be wholly or partially run at the server device **502**. In such a case, the obtaining of an audio stream is by the audio stream modification software **512** running at the server device **502** from the client application **510** running at the participant device **502**, rather than by the audio stream modification software **512** running at the participant device **504** from the client application **510**. Similarly, in such a case, the causing of an output of the modified audio stream within the conference describes the audio stream modification software **512** running at the server device **502** causing a participant device which receives the modified audio stream (e.g., one of the other participant devices **506**) to output the modified audio stream during the conference, rather than the audio stream modification software **512** running at the participant device **504** from which the audio stream derived causing the output based on the client application **510** transmitting the modified audio stream to the server device **502** for distribution by the conferencing software **508**.

(89) FIG. 7 is a block diagram of an example of a system **700** for filtering sensitive topic speech within a conference audio stream. The system **700** includes a first participant device **702**, a second participant device **704**, a third participant device **706**, and a server device **708**. The server device **708** runs conferencing software **710** for implementing an audio or video conference. The participant devices **702** through **706** connect to the conference implemented by the conferencing software **710** respectively using a client application **712**, a client application **714**, and a client application **716**. The first participant device **702** may, for example, be the participant device **504**, in which case the client application **712** may be the client application **510**. The second participant device **704** and the third participant device **706** may, for example, be the other participant devices **506**. The server device **708** may, for example, be the server device **502**, in which case the conferencing software **710** may be the conferencing software **508**.

(90) The participants to the conference are associated with multiple domains. In particular, as shown, the first participant device **702** has a user account **718** associated with a first domain, the second participant device **704** has a user account **720** also associated with the domain, and the third participant device **706** has a user account **722** associated with a second domain. The first participant device **702** and the second participant device **704**, both being associated with the first

domain, have a predefined sensitive topic data store stored thereat. In particular, a data store **724** is stored at the first participant device **702** and a data store **726** is stored at the second participant device **704**. The data store **724** and the data store **726** are identical and may, for example, be copies of a predefined sensitive topic data store maintained by an administrator associated with the first domain. For example, the administrator may cause changes to the predefined sensitive topic data store at their own device to be pushed as updates to the data store **724** and to the data store **726**.

(91) During the conference to which each of the first participant device **702**, the second participant device **704**, and the third participant device **706** are connected, the client application **712** (e.g., using audio stream modification software running thereat, such as the audio stream modification software **512**) processes an audio stream from the first participant device **702** (e.g., produced based on input captured using a microphone of the first participant device **702**) to determine to modify the audio stream to filter sensitive topic speech therein. In particular, based on a determination that the third participant device **706** corresponds to the second domain and thus not the first domain to which the first participant device **702** corresponds, the client application **712** hashes portions of the speech represented within the audio stream and compares those resulting hash values against hash values for records of the data store **724**. For example, the client application **712** may have direct visibility into domains associated with participant devices connected to the conference. In another example, indications of domains with which the participant devices are associated may be transmitted to the client application **712** from the conferencing software **710**. Upon a match being determined based on that comparison, the client application **712** modifies the audio stream to filter the predefined sensitive topic determined to correspond to a portion of the speech. The second participant device **704** and the third participant device **706** obtain the modified audio speech, which is output at those devices.

(92) However, where the third participant device **706** is omitted and thus not connected to the conference, a determination may instead be made by the client application **712** (e.g., by the audio stream modification software running thereat) that all participant devices connected to the conference are of the first domain. In such a case, the client application **712** may forego modifying the audio stream from the first participant device **702**, notwithstanding that the audio stream may include speech which corresponds to a sensitive topic. This is because the exposure of that sensitive topic to other participant device users also of the same domain as the user of the first participant device **702** would not pose a risk with respect to the sensitive topic being disclosed outside of a common entity.

(93) While the audio stream modification functionality of the system **700** is described with respect to the client application **712** and thus as participant device-side functionality, in some implementations, the audio stream modification functionality of the system **700** may be server-side functionality and thus present at the server device **708**. For example, the server device **708** may include audio stream modification software (e.g., as part of the conferencing software **710**) which obtains audio streams from each of the first participant device **702**, the second participant device **704**, and the third participant device **706** and which produces and causes outputs of a modified audio stream at those participant devices as described above.

(94) FIG. **8** is a block diagram of an example of a system **800** for removing undesirable speech from within a conference audio stream. The system **800** includes a first participant device **802**, a second participant device **804**, a third participant device **806**, and a server device **808**. The server device **808** runs conferencing software **810** for implementing an audio or video conference. The participant devices **802** through **806** connect to the conference implemented by the conferencing software **810** respectively using a client application **812**, a client application **814**, and a client application **816**. The first participant device **802** may, for example, be the participant device **504**, in which case the client application **812** may be the client application **510**. The second participant device **804** and the third participant device **806** may, for example, be the other participant devices **506**. The server device **808** may, for example, be the server device **502**, in which case the

conferencing software **810** may be the conferencing software **508**.

(95) The participants to the conference are associated with multiple domains. In particular, as shown and as with the system **700**, the first participant device **802** has a user account **818** associated with a first domain, the second participant device **804** has a user account **820** also associated with the domain, and the third participant device **806** has a user account **822** associated with a second domain. The first participant device **802** and the second participant device **804** each has an audio profile stored thereat. In particular, the first participant device **802** includes or otherwise accesses a first audio profile **824** and the second participant device **804** includes or otherwise accesses a second audio profile **826**. The first audio profile **824** is specific to the user of the first participant device **802** and thus includes elements of language used by the user of the first participant device **802**. The second audio profile **826** is specific to the user of the second participant device **804** and thus includes elements of language used by the user of the second participant device **804**. For example, the user of the first participant device **802** may have control over the contents of the first audio profile **824** and the user of the second participant device **804** may have control over the contents of the second audio profile **826**.

(96) During the conference to which each of the first participant device **802**, the second participant device **804**, and the third participant device **806** are connected, the client application **812** (e.g., using audio stream modification software running thereat, such as the audio stream modification software **512**) processes an audio stream from the first participant device **802** (e.g., produced based on input captured using a microphone of the first participant device **802**) to determine to modify the audio stream to remove undesirable language therefrom. In particular, based on a determination that the third participant device **806** corresponds to the second domain and thus not the first domain to which the first participant device **802** corresponds, the client application **812** hashes portions of the speech represented within the audio stream and compares those resulting hash values against hash values for elements of language within the first audio profile **824**. For example, and as with the system **700**, the client application **812** may have direct visibility into domains associated with participant devices connected to the conference, or indications of domains with which the participant devices are associated may be transmitted to the client application **812** from the conferencing software **810**. Upon a match being determined based on that comparison such that a portion of the speech represented within the audio stream is determined to correspond to language identified as undesirable within the first audio profile **824**, the client application **812** modifies the audio stream to remove the undesirable language determined to correspond to a portion of the speech. The second participant device **804** and the third participant device **806** obtain the modified audio speech, which is output at those devices.

(97) However, where the third participant device **806** is omitted and thus not connected to the conference, a determination may instead be made by the client application **812** (e.g., by the audio stream modification software running thereat) that all participant devices connected to the conference are of the first domain. In such a case, the client application **812** may forego modifying the audio stream from the first participant device **802**, notwithstanding that the audio stream may include speech which corresponds to language identified as undesirable within the first audio profile **824**. This is because the exposure of that undesirable language to other participant device users also of the same domain as the user of the first participant device **702** would not pose a same risk of harm as may result from the exposure of that undesirable language to a participant device user outside of that same domain.

(98) While the audio stream modification functionality of the system **800** is described with respect to the client application **812** and thus as participant device-side functionality, in some implementations, the audio stream modification functionality of the system **800** may be server-side functionality and thus be present at the server device **808**. For example, the server device **808** may include audio stream modification software (e.g., as part of the conferencing software **810**) which obtains audio streams from each of the first participant device **802**, the second participant device

804, and the third participant device **806** and which produces and causes outputs of a modified audio stream at those participant devices as described above.

(99) To further describe some implementations in greater detail, reference is next made to examples of techniques which may be performed by or using a conference participant audio stream modification system. FIG. **9** is a flowchart of an example of a technique **900** for filtering sensitive topic speech within a conference audio stream. FIG. **10** is a flowchart of an example of a technique **1000** for removing undesirable speech from within a conference audio stream.

(100) The technique **900** and/or the technique **1000** can be executed using computing devices, such as the systems, hardware, and software described with respect to FIGS. **1-8**. In one example, the technique **900** may be performed at a server device, such as the server device **502**. In another example, the technique **900** may be performed at a client device, such as the participant device **504**. The technique **900** and/or the technique **1000** can be performed, for example, by executing a machine-readable program or other computer-executable instructions, such as routines, instructions, programs, or other code. The steps, or operations, of the technique **900** and/or the technique **1000**, or another technique, method, process, or algorithm described in connection with the implementations disclosed herein can be implemented directly in hardware, firmware, software executed by hardware, circuitry, or a combination thereof.

(101) For simplicity of explanation, the technique **900** and the technique **1000** are each depicted and described herein as a series of steps or operations. However, the steps or operations of the technique **900** in accordance with this disclosure can occur in various orders and/or concurrently. Additionally, other steps or operations not presented and described herein may be used. Furthermore, not all illustrated steps or operations may be required to implement a technique in accordance with the disclosed subject matter.

(102) Referring first to FIG. **9**, the technique **900** for filtering sensitive topic speech within a conference audio stream is shown. At **902**, an audio stream is obtained from a participant device connected to a conference. The audio stream represents speech of a user of the participant device. The conference may, for example, be an audio conference or a video conference. In at least some cases, the conference may be implemented by a UCaaS or other software platform.

(103) At **904**, hash values are determined for portions of the speech. The hash values determined for the portions of the speech, referred to as first hash values, are determined on a temporal basis (e.g., a time chunk of the audio stream) or spatial basis (e.g., for a single word, a combination of words, or a complete clause, phrase, or sentence). Determining the first hash values includes computing the hash values based on the content of the portions of the speech, for example, using a hashing algorithm such as SHA-1, SHA-2, or SHA-3.

(104) At **906**, a determination is made based on a hash value comparison that a portion of the speech corresponds to a predefined sensitive topic. In particular, the first hash values are compared against second hash values for records of a data store, such as a predefined sensitive topic data store, to determine whether there is a match between any of the first hash values and any of the second hash values. For example, the records of the data store may be populated by an entity having a domain with which a conference user account used at the participant device is associated. Each record of the data store may indicate a different predefined sensitive topic. A determination is made that a portion of the speech corresponds to a predefined sensitive topic based on a first hash value corresponding to that portion of the speech matching a second hash value corresponding to a record for that predefined sensitive topic within the predefined sensitive topic data store. In non-limiting examples, the predefined sensitive topic may relate to confidential information or a codename.

(105) At **908**, a filter is applied against the portion of the speech corresponding to the matching hash value to produce a modified audio stream within which the predefined sensitive topic is sanitized (e.g., omitted, obfuscated, or otherwise replaced). The filter includes data or other configurations for modifying the audio stream, specifically, the portion of the speech within the

audio stream. In some cases, the application of the filter causes an omission of the portion of the speech from within the modified audio stream. In other cases, the application of the filter causes an obfuscation of the portion of the speech from within the modified audio stream. In still other cases, the application of the filter causes a replacement of the portion of the speech from within the modified audio stream with other audible content. Portions of the speech other than the portion of the speech remain unfiltered within the modified audio stream.

(106) At **910**, the output of the modified audio stream is caused in place of the audio stream (i.e., as originally obtained from the participant device above). For example, causing the output of the modified audio stream in place of the audio stream can include causing the modified audio stream to be output at one or more participant devices connected to the conference based on conference user accounts used at the one or more participant devices corresponding other than to a domain associated with a conference user account used at the participant device from which the audio stream is obtained. In another example, causing the output of the modified audio stream in place of the audio stream can include transmitting the modified audio stream to one or more participant devices connected to the conference to cause the modified audio stream to be output at the one or more participant devices. Thus, the modified audio stream is output at all participant devices connected to the conference. Alternatively, the modified audio stream may be output at a limited number of participant devices connected to the conference. In one example, the modified audio stream may be output only at participant devices other than the one from which the audio stream is obtained. In another example, the modified audio stream may be output only at a subset of participant devices connected to the conference based on user accounts used at the subset of participant devices to access the conference having a domain that is different from a domain of the user account used at the participant device from which the audio stream is obtained.

(107) In some implementations, the technique **900** can include determining to compare the first hash values against the second hash values based on a mismatch between a first domain associated with a first conference user account used at the participant device and a second domain associated with a second conference user account used at another participant device connected to the conference. In some implementations, the technique **900** can include determining to apply the filter against the portion of the speech based on a mismatch between a first domain associated with a first conference user account used at the participant device and a second domain associated with a second conference user account used at another participant device connected to the conference. In some implementations, a control policy restricts users of participant devices connected to the conference from altering the records of the data store. In some implementations, a record of the data store is removed from the data store upon a public disclosure of an associated predefined sensitive topic.

(108) In some implementations, the technique **900** may include converting the portion of the speech into text and comparing the text against text of records of a predefined sensitive topic data store to determine that the portion of the speech corresponds to a predefined sensitive topic. In at least some such cases, operations related to hashing may be omitted from the technique **900**.

(109) Referring next to FIG. **10**, the technique **1000** for removing undesirable speech within a conference audio stream is shown. At **1002**, an audio stream is obtained from a first participant device connected to a conference. The audio stream represents speech of a user of the participant device. The conference may, for example, be an audio conference or a video conference. In at least some cases, the conference may be implemented by a UCaaS or other software platform.

(110) At **1004**, a portion of the speech is detected as corresponding to language identified as undesirable within an audio profile. In some cases, the audio profile corresponds to the user of the participant device and identifies, as undesirable, a plurality of language spoken by the user of the participant device in one or more past conferences. For example, in some such cases, the audio profile is configured by the user to identify select language as undesirable. In some cases, the audio profile corresponds to an entity with which the user of the participant is associated and identifies,

as undesirable, a plurality of language spoken by one or more participant device users associated with the entity in one or more past conferences. The audio profile may in at least some cases be stored at the participant device from which the audio stream is obtained. The audio profile may be updated by an administrator of a user account of the user of the participant device.

(111) Detecting that the portion of the speech corresponds to language identified as undesirable within the audio profile can include determining a hash value for the portion of the speech and comparing the hash value against hash values of the audio profile. For example, hash values may be determined for portions of the speech. The hash values determined for the portions of the speech, referred to as first hash values, may be determined on a temporal basis (e.g., a time chunk of the audio stream) or spatial basis (e.g., for a single word, a combination of words, or a complete clause, phrase, or sentence). Determining the first hash values includes computing the hash values based on the content of the portions of the speech, for example, using a hashing algorithm such as SHA-1, SHA-2, or SHA-3. The first hash values are compared against second hash values for elements of language within the audio profile to determine whether there is a match between any of the first hash values and any of the second hash values. A determination is made that a portion of the speech corresponds to language identified as undesirable based on a first hash value corresponding to that portion of the speech matching a second hash value corresponding to an element of language within the audio profile. Alternatively, where the technique **1000** is performed at a server device, detecting that the portion of the speech corresponds to language identified as undesirable can include obtaining, from a client application running at the participant device, an indication that the portion of the speech corresponds to the language identified as undesirable. In non-limiting examples, the portion of the speech determined as including undesirable language may include profanity or a filler word.

(112) At **1006**, a modified audio stream is produced by removing the portion of the speech from the audio stream. For example, removing the portion of the speech from the audio stream can include obfuscating the portion of the speech within the modified audio stream, omitting the portion of the speech from the modified audio stream, or replacing the portion of the speech with something else within the modified audio stream. Portions of the speech other than the portion of the speech may remain unmodified and unremoved within the modified audio stream.

(113) At **1008**, the output of the modified audio stream is caused in place of the audio stream (i.e., as originally obtained from the participant device above). The modified audio stream is output at all participant devices connected to the conference. Alternatively, the modified audio stream may be output at a limited number of participant devices connected to the conference. In one example, the modified audio stream may be output only at participant devices other than the one from which the audio stream is obtained. In another example, the modified audio stream may be output only at a subset of participant devices connected to the conference based on user accounts used at the subset of participant devices to access the conference having a domain that is different from a domain of the user account used at the participant device from which the audio stream is obtained.

(114) In some implementations, the technique **1000** can include applying a filter to automate a removal of further instances of the language identified as undesirable from the audio stream. In some implementations, the technique **1000** can include presenting, based on detecting that the portion of the speech corresponds to language identified as undesirable, a prompt to the user of a participant device recommending to assert a filter against future audio stream content obtained from the participant device. In some implementations, the technique **1000** can include storing, based on the modification of the portion of the speech, configuration data indicating to remove the language identified as undesirable from a future audio stream during one or more future conferences. In some implementations, the technique **1000** can include notifying the user of the participant device of the removal of the portion of the speech from the audio stream.

(115) In some implementations, the technique **1000** may include converting the portion of the speech into text and comparing the text against text of an audio profile to determine that the portion

of the speech corresponds to language identified as undesirable. In at least some such cases, operations related to hashing may be omitted from the technique **1000**.

(116) The implementations of this disclosure correspond to methods, non-transitory computer readable media, apparatuses, systems, devices, and the like. In some implementations, a method comprises obtaining an audio stream representing speech of a user of a participant device connected to a conference; determining first hash values for portions of the speech; determining, by comparing the first hash values against second hash values for records of a data store, that a portion of the speech corresponds to a predefined sensitive topic indicated within the records; applying a filter against the portion of the speech to produce a modified audio stream within which the predefined sensitive topic is sanitized; and causing an output, within the conference, of the modified audio stream in place of the audio stream. In some implementations, a non-transitory computer readable medium stores instructions operable to cause one or more processors to perform operations comprising obtaining an audio stream representing speech of a user of a participant device connected to a conference; determining first hash values for portions of the speech; determining, by comparing the first hash values against second hash values for records of a data store, that a portion of the speech corresponds to a predefined sensitive topic indicated within the records; applying a filter against the portion of the speech to produce a modified audio stream within which the predefined sensitive topic is sanitized; and causing an output, within the conference, of the modified audio stream in place of the audio stream. In some implementations, an apparatus comprises a memory and a processor configured to execute instructions stored in the memory to obtain an audio stream representing speech of a user of a participant device connected to a conference; determine first hash values for portions of the speech; determine, by comparing the first hash values against second hash values for records of a data store, that a portion of the speech corresponds to a predefined sensitive topic indicated within the records; apply a filter against the portion of the speech to produce a modified audio stream within which the predefined sensitive topic is sanitized; and cause an output, within the conference, of the modified audio stream in place of the audio stream.

(117) In some implementations of the method, non-transitory computer readable medium, or apparatus, portions of the speech other than the portion of the speech remain unfiltered within the modified audio stream.

(118) In some implementations of the method, non-transitory computer readable medium, or apparatus, the records of the data store are populated by an entity having a domain with which a conference user account used at the participant device is associated.

(119) In some implementations of the method, non-transitory computer readable medium, or apparatus, each record of the data store indicates a different predefined sensitive topic, and a record of the data store is removed from the data store upon a public disclosure of an associated predefined sensitive topic.

(120) In some implementations of the method, non-transitory computer readable medium, or apparatus, the predefined sensitive topic relates to confidential information.

(121) In some implementations of the method, non-transitory computer readable medium, or apparatus, the application of the filter causes an omission of the portion of the speech from within the modified audio stream.

(122) In some implementations of the method, non-transitory computer readable medium, or apparatus, the application of the filter causes an obfuscation of the portion of the speech from within the modified audio stream.

(123) In some implementations of the method, non-transitory computer readable medium, or apparatus, the application of the filter causes a replacement of the portion of the speech from within the modified audio stream with other audible content.

(124) In some implementations of the method, non-transitory computer readable medium, or apparatus, causing the output of the modified audio stream in place of the audio stream comprises:

causing the modified audio stream to be output at one or more participant devices connected to the conference based on conference user accounts used at the one or more participant devices corresponding other than to a domain associated with a conference user account used at the participant device from which the audio stream is obtained.

(125) In some implementations of the method, non-transitory computer readable medium, or apparatus, causing the output of the modified audio stream in place of the audio stream comprises: transmitting the modified audio stream to one or more participant devices connected to the conference to cause the modified audio stream to be output at the one or more participant devices.

(126) In some implementations of the method, non-transitory computer readable medium, or apparatus, the predefined sensitive topic corresponds to a codename.

(127) In some implementations of the method, non-transitory computer readable medium, or apparatus, the conference is implemented by a UCaaS software platform.

(128) In some implementations of the method, non-transitory computer readable medium, or apparatus, the method comprises, the operations comprise, and the processor is configured to execute the instructions for determining to compare the first hash values against the second hash values based on a mismatch between a first domain associated with a first conference user account used at the participant device and a second domain associated with a second conference user account used at another participant device connected to the conference.

(129) In some implementations of the method, non-transitory computer readable medium, or apparatus, the method comprises, the operations comprise, and the processor is configured to execute the instructions for determining to apply the filter against the portion of the speech based on a mismatch between a first domain associated with a first conference user account used at the participant device and a second domain associated with a second conference user account used at another participant device connected to the conference.

(130) In some implementations of the method, non-transitory computer readable medium, or apparatus, the modified audio stream is output to a first subset of participant devices connected to the conference and the audio stream is output to a second subset of the participant devices, conference user accounts used at the first subset of the participant devices are associated with a first domain different from a second domain with which a conference user account used at the participant device is associated with, and conference user accounts used at the second subset of the participant devices are associated with the second domain.

(131) In some implementations of the method, non-transitory computer readable medium, or apparatus, a control policy restricts users of participant devices connected to the conference from altering the records of the data store.

(132) In some implementations of the method, non-transitory computer readable medium, or apparatus, the modified audio stream is produced and output while the conference remains ongoing.

(133) In some implementations of the method, non-transitory computer readable medium, or apparatus, the modified audio stream is produced and output during playback of a recording of the conference.

(134) The implementations of this disclosure can be described in terms of functional block components and various processing operations. Such functional block components can be realized by a number of hardware or software components that perform the specified functions. For example, the disclosed implementations can employ various integrated circuit components (e.g., memory elements, processing elements, logic elements, look-up tables, and the like), which can carry out a variety of functions under the control of one or more microprocessors or other control devices. Similarly, where the elements of the disclosed implementations are implemented using software programming or software elements, the systems and techniques can be implemented with a programming or scripting language, such as C, C++, Java, JavaScript, assembler, or the like, with the various algorithms being implemented with a combination of data structures, objects, processes,

routines, or other programming elements.

(135) Functional aspects can be implemented in algorithms that execute on one or more processors. Furthermore, the implementations of the systems and techniques disclosed herein could employ a number of conventional techniques for electronics configuration, signal processing or control, data processing, and the like. The words “mechanism” and “component” are used broadly and are not limited to mechanical or physical implementations, but can include software routines in conjunction with processors, etc. Likewise, the terms “system” or “tool” as used herein and in the figures, but in any event based on their context, may be understood as corresponding to a functional unit implemented using software, hardware (e.g., an integrated circuit, such as an ASIC), or a combination of software and hardware. In certain contexts, such systems or mechanisms may be understood to be a processor-implemented software system or processor-implemented software mechanism that is part of or callable by an executable program, which may itself be wholly or partly composed of such linked systems or mechanisms.

(136) Implementations or portions of implementations of the above disclosure can take the form of a computer program product accessible from, for example, a computer-usable or computer-readable medium. A computer-usable or computer-readable medium can be a device that can, for example, tangibly contain, store, communicate, or transport a program or data structure for use by or in connection with a processor. The medium can be, for example, an electronic, magnetic, optical, electromagnetic, or semiconductor device.

(137) Other suitable mediums are also available. Such computer-usable or computer-readable media can be referred to as non-transitory memory or media, and can include volatile memory or non-volatile memory that can change over time. The quality of memory or media being non-transitory refers to such memory or media storing data for some period of time or otherwise based on device power or a device power cycle. A memory of an apparatus described herein, unless otherwise specified, does not have to be physically contained by the apparatus, but is one that can be accessed remotely by the apparatus, and does not have to be contiguous with other memory that might be physically contained by the apparatus.

(138) While the disclosure has been described in connection with certain implementations, it is to be understood that the disclosure is not to be limited to the disclosed implementations but, on the contrary, is intended to cover various modifications and equivalent arrangements included within the scope of the appended claims, which scope is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures as is permitted under the law.

Claims

1. A method, comprising: obtaining an audio stream representing speech of a user of a participant device connected to a conference; determining first hash values for portions of the speech; determining, by comparing the first hash values against second hash values for records of a data store, that a portion of the speech corresponds to a predefined sensitive topic indicated within the records; applying a filter against the portion of the speech to produce a modified audio stream within which the predefined sensitive topic is sanitized; and causing an output, within the conference, of the modified audio stream in place of the audio stream.
2. The method of claim 1, wherein portions of the speech other than the portion of the speech remain unfiltered within the modified audio stream.
3. The method of claim 1, wherein the records of the data store are populated by an entity having a domain with which a conference user account used at the participant device is associated.
4. The method of claim 1, wherein each record of the data store indicates a different predefined sensitive topic, and wherein a record of the data store is removed from the data store upon a public disclosure of an associated predefined sensitive topic.
5. The method of claim 1, wherein the predefined sensitive topic relates to confidential

information.

6. The method of claim 1, wherein the application of the filter causes an omission of the portion of the speech from within the modified audio stream.

7. The method of claim 1, wherein the application of the filter causes an obfuscation of the portion of the speech from within the modified audio stream.

8. The method of claim 1, wherein the application of the filter causes a replacement of the portion of the speech from within the modified audio stream with other audible content.

9. A non-transitory computer readable medium storing instructions operable to cause one or more processors to perform operations comprising: obtaining an audio stream representing speech of a user of a participant device connected to a conference; determining first hash values for portions of the speech; determining, by comparing the first hash values against second hash values for records of a data store, that a portion of the speech corresponds to a predefined sensitive topic indicated within the records; applying a filter against the portion of the speech to produce a modified audio stream within which the predefined sensitive topic is sanitized; and causing an output, within the conference, of the modified audio stream in place of the audio stream.

10. The non-transitory computer readable medium of claim 9, wherein the operations for causing the output of the modified audio stream in place of the audio stream comprise: causing the modified audio stream to be output at one or more participant devices connected to the conference based on conference user accounts used at the one or more participant devices corresponding other than to a domain associated with a conference user account used at the participant device from which the audio stream is obtained.

11. The non-transitory computer readable medium of claim 9, wherein the operations for causing the output of the modified audio stream in place of the audio stream comprise: transmitting the modified audio stream to one or more participant devices connected to the conference to cause the modified audio stream to be output at the one or more participant devices.

12. The non-transitory computer readable medium of claim 9, wherein the predefined sensitive topic corresponds to a codename.

13. The non-transitory computer readable medium of claim 9, wherein the conference is implemented by a unified communications as a service software platform.

14. An apparatus, comprising: a memory; and a processor configured to execute instructions stored in the memory to: obtain an audio stream representing speech of a user of a participant device connected to a conference; determine first hash values for portions of the speech; determine, by comparing the first hash values against second hash values for records of a data store, that a portion of the speech corresponds to a predefined sensitive topic indicated within the records; apply a filter against the portion of the speech to produce a modified audio stream within which the predefined sensitive topic is sanitized; and cause an output, within the conference, of the modified audio stream in place of the audio stream.

15. The apparatus of claim 14, wherein the processor is configured to execute the instructions to: determine to compare the first hash values against the second hash values based on a mismatch between a first domain associated with a first conference user account used at the participant device and a second domain associated with a second conference user account used at another participant device connected to the conference.

16. The apparatus of claim 14, wherein the processor is configured to execute the instructions to: determine to apply the filter against the portion of the speech based on a mismatch between a first domain associated with a first conference user account used at the participant device and a second domain associated with a second conference user account used at another participant device connected to the conference.

17. The apparatus of claim 14, wherein the modified audio stream is output to a first subset of participant devices connected to the conference and the audio stream is output to a second subset of the participant devices, wherein conference user accounts used at the first subset of the participant

devices are associated with a first domain different from a second domain with which a conference user account used at the participant device is associated with, and wherein conference user accounts used at the second subset of the participant devices are associated with the second domain.

18. The apparatus of claim 14, wherein a control policy restricts users of participant devices connected to the conference from altering the records of the data store.

19. The apparatus of claim 14, wherein the modified audio stream is produced and output while the conference remains ongoing.

20. The apparatus of claim 14, wherein the modified audio stream is produced and output during playback of a recording of the conference.
